Experiment ‘p_6_2_j_has_duplicates_array’ Results

June 4, 2026

Experiment outcome: SUCCESS
Bad responses: 0
Responses containing `assume` **:** 0
Responses containing `expect` **:** 0
Resolution attempts: 1
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 1

Problem Specification

Problem name: p_6_2_j_has_duplicates_array
Natural language statement: Write a method to return true if the given array contains duplicate elements (which need not be adjacent).
Method signature: p_6_2_j_has_duplicates_array(inputs: array<int>) returns (has_duplicates: bool)

Ensures

- `has_duplicates == (exists i, j :: 0 <= i < j < inputs.Length && inputs[i] == inputs[j])`

Requires

- `inputs.Length >= 2`

GenAI interactions

Below you will find all interactions between the 'user' (program) and the 'assistant' (OpenAI).

Program → GenAI

```
You are given the following task to perform in Dafny:

Write a method to return true if the given array contains duplicate elements (which need
    ↪ not be adjacent).

The signature should be:

p_6_2_j_has_duplicates_array(inputs: array<int>) returns (has_duplicates: bool)

The method should respect the following contract:

requires inputs.Length >= 2, ensures has_duplicates == (exists i, j :: 0 <= i < j <
    ↪ inputs.Length && inputs[i] == inputs[j])

Produce and show only the Dafny body of this method, including the curly braces that
    ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
    ↪  'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-Dn1TflbIYi73IvSAytRvqamjGFRL5
Model: gpt-5.5-2026-04-23
Created at: 1780576123
Finish reason: stop
Usage: 326 tokens in, and 3150 tokens out

```
{
  has_duplicates := exists i, j | 0 <= i < j < inputs.Length :: inputs[i] == inputs[j];
}
```

Final Program

```
method p_6_2_j_has_duplicates_array(inputs: array<int>) returns (has_duplicates: bool)
       requires inputs.Length >= 2
       ensures has_duplicates == (exists i, j :: 0 <= i < j < inputs.Length && inputs[i]
           ↪ == inputs[j])
{
  has_duplicates := exists i, j | 0 <= i < j < inputs.Length :: inputs[i] == inputs[j];
}
```

Total Token Usage

Input tokens: 326
Output tokens: 3150
Reasoning tokens: 3072
Sum of 'total tokens': 3476

Experiment Timings

Overall Experiment started at 1780576122680, ended at 1780576219939, lasting 97259ms (97.26 seconds)
Iteration #1 started at 1780576122680, ended at 1780576219939, lasting 97259ms (97.26 seconds)